

Mechanism identifiability in connectome-based neurodegeneration: decoupling intracellular seeding from trans-synaptic spread reveals topology-invariant and topology-dependent disease entry points

Maryam Rezaee

Independent Researcher

maryamrezaeenl@gmail.com

June 2026

CRITICAL DISCLAIMERS: This is a computational modelling study using the *C. elegans* 61-neuron motor connectome as a substrate for studying abstract degeneration dynamics. It is **not a clinical model of ALS**, does not use human neural circuitry, and cannot make predictions about human patients. The *C. elegans* motor system differs fundamentally from the human corticospinal-motor neuron system affected in ALS. All model parameters are chosen for dynamical plausibility, not biological calibration to measured quantities in any organism. **All results are hypothesis-generating and require experimental validation before any translational significance can be attributed to them.**

Abstract

Connectome-based neurodegeneration models typically aggregate protein seeding and synaptic spread into a single amplification parameter, leaving the relative causal roles of cell-autonomous and circuit-dependent mechanisms unresolved. We address this identifiability problem by extending a previously validated *C. elegans* motor-circuit cascade model [Rezaee, 2026] to decouple intracellular seeding rate (ISR) from trans-synaptic spread efficiency (TSSE).

Across a 5×5 parameter grid (25 cells, 500 baseline runs), both mechanisms independently sustain genuine triphasic tipping (mean rate 83.8%), with ISR more predictive of tipping probability (Pearson $r = 0.586$ vs. 0.463 for TSSE). A single-mechanism ablation protocol applied across six cascade pathways and three dynamical regimes confirms that ISR and TSSE are the sole universally load-bearing mechanisms; mitochondrial damage acquires load-bearing character only under extreme fragility ($\text{mitFrag} \geq 4.0$), while glutamate excitotoxicity, calcium/ROS, and irreversibility remain negligible across all regimes.

Topology replication across five graph types (8100 runs) reveals a decisive mechanistic dissociation: ISR-dominant cascades achieve 100% genuine tipping on

all five topologies, whereas TSSE-dominant cascades succeed only on the biological *C. elegans* connectome (100%) and fail on all four synthetic alternatives (0–5%). A vulnerability-alignment experiment (R3.7) shows that BA topology’s resistance reflects hub-vulnerability misalignment rather than hub architecture per se. A seed-location sweep of all 61 neurons confirms seed location as a timing modulator (≤ 88 -step window) rather than a tipping determinant (genuine rate = 1.000 for 59/61 neurons).

Three therapeutic-mapping phases extend these findings into intervention design. Glutamate suppression is 63-fold less effective than ISR suppression (R3.8). A sharp therapeutic cliff at 85–90% ISR suppression produces a 28-percentage-point single-step drop in genuine tipping rate; $\text{ISR}_{50} = 96.5\%$ (R3.9). Production suppression (ASO analogue) substantially outperforms spread inhibition at all tested doses; coupled co-suppression is synergistic above 60% strength by Bliss Independence analysis (synergy +0.47 at 80%, CI [+0.38, +0.53]), with synergy onset coinciding exactly with the R3.9 therapeutic cliff (R3.10).

These findings sharpen the mechanistic claims of v1.0, generate testable predictions distinguishing cell-autonomous from circuit-dependent ALS pathology, and identify the ISR production pathway as the primary target for cascade prevention.

Keywords: neurodegeneration; connectome; protein aggregation; trans-synaptic spread; tipping points; ALS; *C. elegans*; mechanism identifiability; topology; vulnerability alignment; hub structure

Contents

1	Introduction	4
1.1	Background	4
1.2	The identifiability problem	4
1.3	Round 3 scope	4
2	Methods	5
2.1	Model: decoupled aggregation (v2.0)	5
2.2	Parameter grid (R3.1)	6
2.3	Ablation protocol (R3.2)	6
2.4	Mitochondrial threshold protocol (R3.3/R3.4)	6
2.5	Topology test protocol (R3.5)	6
2.6	Seed location protocol (R3.6)	7
2.7	Vulnerability-alignment protocol (R3.7)	7
2.8	Combination therapy protocol (R3.8)	7
2.9	Therapeutic cliff protocol (R3.9)	7
2.10	Biological mapping protocol (R3.10)	7
3	Results	8
3.1	R3.1 — Decoupled aggregation: ISR and TSSE independently load-bearing	8
3.2	R3.2 — Downstream causal power: mitochondria conditional, others negligible	9
3.3	R3.3 and R3.4 — Mitochondrial threshold: near-takeover at $\text{mitFrag} = 4.0$	10
3.4	R3.5 — Topological necessity: ISR topology-invariant, TSSE topology-sensitive	10
3.5	R3.6 — Seed location: timing modulator, not tipping determinant	11
3.6	R3.7 — Degree-correlated vulnerability: BA failure reflects misalignment	12

3.7	R3.8 — Combination therapy: ISR is cascade gatekeeper, glutamate negligible	13
3.8	R3.9 — Therapeutic cliff: dose-response is threshold-gated	14
3.9	R3.10 — Biological mapping: production suppression dominates; synergy above 60%	16
4	Discussion	17
4.1	ISR as cell-autonomous disease mechanism	17
4.2	TSSE as circuit-dependent disease mechanism	17
4.3	Mitochondria as death accelerator, not cascade gatekeeper	18
4.4	Implications for v1.0 claims	18
4.5	Therapeutic implications of R3.8–R3.10	18
4.6	Limitations of the mechanistic interpretation	19
5	Conclusion	19
6	Limitations	20
	Data Availability	20

1 Introduction

1.1 Background

Computational models of ALS-related neurodegeneration increasingly incorporate network structure as a substrate for propagating pathological protein aggregation [Neumann et al., 2006]. A key modelling challenge is *mechanism identifiability*: when a single parameter controls both local protein accumulation and synaptic propagation, it is impossible to determine which process is the primary driver of network-level collapse. This limits the translational value of parameter sensitivity analyses and ablation studies.

In v1.0 [Rezaee, 2026], we demonstrated that a biophysical cascade (aggregation $\rightarrow$ mitochondrial damage $\rightarrow$ ATP collapse $\rightarrow$ glutamate excitotoxicity $\rightarrow$ calcium/ROS $\rightarrow$ aggregation) running on the empirical *C. elegans* 61-neuron motor connectome produces genuine triphasic tipping points, a sharp linear therapeutic boundary, and two disease subtypes robust to biological noise. The dominant parameter, **aggregationAmplification** (aggAmp), explained 84.6% of subtype membership variance (Cohen’s $d = 51.5$) and separated subtypes with Pearson $r = 0.999$.

However, the v1.0 aggAmp parameter controlled *both* intracellular seeding and trans-synaptic spread through a single multiplicative scalar. This coupling raises a mechanism identifiability question: is aggAmp dominance a property of the underlying biology, or a coupling artifact?

1.2 The identifiability problem

Separating cell-autonomous from circuit-dependent disease mechanisms matters for intervention design. If intracellular seeding is primary, therapies targeting individual neurons (e.g. ASO-mediated TDP-43 stabilisation) may suffice regardless of circuit topology. If trans-synaptic spread is primary, circuit structure becomes a primary determinant of disease progression, and topological interventions become necessary.

The v1.0 topology validation (Phase 7C) showed that Barabási–Albert (BA) scale-free graphs achieved 0% genuine tipping while the *C. elegans* connectome achieved 100%, confirming strong topology dependence. But the coupled aggAmp parameter did not permit distinguishing which sub-mechanism drove this result.

1.3 Round 3 scope

This paper reports Round 3 (R3.1–R3.10), a ten-phase extension addressing mechanism identifiability and its therapeutic implications.

Mechanistic dissection (R3.1–R3.7):

- **R3.1:** Decouple aggAmp into ISR and TSSE; characterise the 5×5 parameter landscape
- **R3.2:** Downstream causal power — can mitochondria, glutamate, or calcium become load-bearing?
- **R3.3/R3.4:** Map and validate the mitochondrial near-takeover threshold with bootstrapped confidence intervals
- **R3.5:** Replicate the topology test with the decoupled model; isolate which mechanism drives topology sensitivity

- **R3.6:** Test seed location sensitivity
- **R3.7:** Determine whether BA’s resistance reflects hub-vulnerability misalignment or hub architecture per se

Therapeutic mapping (R3.8–R3.10):

- **R3.8:** ISR $\times$ glutamate combination therapy sweep
- **R3.9:** ISR suppression dose-response shape (cliff mapping)
- **R3.10:** Biological mapping — production suppression vs. spread inhibition

All phases use the *C. elegans* motor connectome (61 neurons, 127 directed synapses; [White et al. 1986](#); [Cook et al. 2019](#)) and the strict three-part Phase 7B tipping criterion established in v1.0.

2 Methods

2.1 Model: decoupled aggregation (v2.0)

The v1.0 aggregation update equation was:

$$\begin{aligned}
 \Delta a_i &= \text{vuln}_i \cdot \text{SEED_RATE} \cdot \text{aggAmp} \cdot \Delta t && \text{(seeding)} \\
 &+ \text{SPREAD_RATE} \cdot \text{aggAmp} \cdot \sum_{j \in \mathcal{N}(i)} w_{ij} \cdot a_j \cdot \Delta t && \text{(spread)} \\
 &+ \text{oxFeedback} \cdot \text{ox}_i \cdot \Delta t && \text{(oxidative)}
 \end{aligned}$$

where vuln_i is neuron i ’s vulnerability score (DA class = 1.0, DB = 0.90, VA = 0.95, VB = 0.85, DD/VD = 0.65–0.70, interneurons = 0.15–0.30, sensory = 0.05), w_{ij} is the synaptic weight from presynaptic neuron j , and ox_i is oxidative stress.

The v2.0 decoupled equation replaces the single aggAmp scalar with two independent parameters:

$$\begin{aligned}
 \Delta a_i &= \text{vuln}_i \cdot \text{SEED_RATE} \cdot \mathbf{ISR} \cdot \Delta t && \text{(intracellular seeding)} \\
 &+ \text{SPREAD_RATE} \cdot \mathbf{TSSE} \cdot \sum_{j \in \mathcal{N}(i)} w_{ij} \cdot a_j \cdot \Delta t && \text{(trans-synaptic spread)} \\
 &+ \text{oxFeedback} \cdot \text{ox}_i \cdot \Delta t && \text{(oxidative feedback)} \quad (1)
 \end{aligned}$$

ISR (intracellular seeding rate) scales per-neuron intrinsic aggregation growth proportional to vulnerability — a cell-autonomous process independent of synaptic connectivity. **TSSE** (trans-synaptic spread efficiency) scales prion-like propagation along connectome edges — an inherently circuit-dependent mechanism. Backward compatibility: setting $\text{ISR} = \text{TSSE} = \text{aggAmp}$ reproduces v1.0 dynamics exactly. There is no decay term on aggregation; accumulation is bounded only by $\text{clip}(a_i, 0, 1)$.

2.2 Parameter grid (R3.1)

We swept $\text{ISR} \in \{0.05, 0.5, 2.0, 5.0, 10.0\} \times \text{TSSE} \in \{0.05, 0.5, 2.0, 5.0, 10.0\}$, yielding 25 cells. Each cell: 20 random seeds $\times$ 500 steps = 500 baseline runs. Ablation runs (10 seeds, each mechanism disabled to 0.001) confirmed load-bearing status. **Medium context** ($\text{ISR} = \text{TSSE} = 2.0$) was selected for topology and seed-location tests (100% genuine tipping at this diagonal point).

2.3 Ablation protocol (R3.2)

Six mechanisms were ablated one at a time:

Table 1: Ablation protocol: mechanism, code, and parameter setting.

Code	Mechanism	Parameter setting
A	Intracellular seeding	$\text{ISR} = 0.001$
B	Trans-synaptic spread	$\text{TSSE} = 0.001$
C	Mitochondrial damage	$\text{mitFrag} = 0.001$
D	Glutamate excitotoxicity	$\text{glutSens} = 1 \times 10^{-6}$
E	Calcium / ROS	$\text{calcGain} = 0.001$
F	Irreversibility lock	$\text{recIrrev} = 0.999$

Three dynamical regimes were tested: seeding-dominant (ISR high, TSSE moderate), spread-dominant (ISR low, TSSE high), and downstream-stressed (moderate ISR/TSSE, elevated $\text{mitFrag}/\text{glutSens}/\text{calcGain}$). Effect size: *large* = first-death shift > 50 steps OR plateau gain > 10 neurons OR genuine-rate drop > 0.30 .

2.4 Mitochondrial threshold protocol (R3.3/R3.4)

R3.3 swept $\text{mitFrag} \in \{0.3, 0.5, 1.0, 1.5, 2.0, 3.0, 4.0, 5.0, 6.0, 8.0\}$ across three aggregation contexts (low: $\text{ISR}=\text{TSSE}=0.5$; medium: 2.0; high: 5.0), $n = 15$ seeds per cell (900 runs). R3.4 validated the low context with $n = 50$ seeds and 10 000 percentile-bootstrap resamples for 95% CIs on shift, plateau gain, and genuine-rate drop. Near-takeover required *all three*: shift > 50 steps AND gain > 5 neurons AND $\text{rdrop} > 0.20$.

2.5 Topology test protocol (R3.5)

Five graph types at medium context ($\text{ISR} = \text{TSSE} = 2.0$):

1. *C. elegans* biological connectome (1 instance)
2. Erdős–Rényi random directed graph (20 instances, $n = 61$, $m = 127$)
3. Barabási–Albert scale-free (20 instances)
4. Watts–Strogatz small-world (20 instances, $k = 4$, $p = 0.3$)
5. Degree-preserved double-edge-swap shuffle (20 instances)

Mechanism-isolation configurations: ISR-dominant ($\text{ISR} = 5.0$, $\text{TSSE} = 0.5$); spread-dominant ($\text{ISR} = 0.5$, $\text{TSSE} = 5.0$). Total: 10 parameter configs $\times$ 10 seeds $\times$ 81 topology instances = 8 100 runs $\times$ 500 steps.

2.6 Seed location protocol (R3.6)

Initial aggregation boost of +0.5 applied to each of the 61 neurons in turn (10 seeds $\times$ 10 parameter configurations = 100 runs per neuron, 6 100 runs total). Three named conditions: AVAL boost (highest-betweenness command interneuron), DA6 boost (strongest protective node), and uniform baseline (zero initial aggregation). Seed-location sensitivity measured separately for *whether* tipping occurs (genuine tipping rate) and *when* (first death step). AVAL was used as proxy for DVA (absent from the 61-neuron circuit).

2.7 Vulnerability-alignment protocol (R3.7)

Three vulnerability assignments applied to 20 BA instances and the *C. elegans* connectome, under TSSE-dominant context (ISR = 0.5, TSSE = 5.0): (A) original biological gradient; (B) degree-correlated: $\text{vuln}_i = 0.1 + 0.9 \times (d_i/d_{\max})$; (C) inverse-degree: $\text{vuln}_i = 1.0 - 0.9 \times (d_i/d_{\max})$. Total: 630 runs $\times$ 500 steps.

2.8 Combination therapy protocol (R3.8)

A 5 $\times$ 4 grid of ISR suppression $\times$ glutamate suppression under medium context. ISR reductions: {0%, 10%, 20%, 30%, 50%}; glutamate reductions: {0%, 50%, 70%, 90%}; additional solo conditions: ISR 70% only; glutamate 99% only. Total: 22 conditions $\times$ 20 seeds $\times$ 500 steps = 440 runs. Benefit score B : weighted composite of tipping prevention ($w = 0.40$), first-death delay (0.20), plateau survivor gain (0.25), and functional survival gain (0.15), each clamped $[0, 1]$ against untreated baseline. Synergy classified as synergistic ($> +0.10$, $\text{CI}_{\text{lo}} > 0$), antagonistic (< -0.10 , $\text{CI}_{\text{hi}} < 0$), or additive. Bootstrap CI: $N = 500$ resamples.

Note: clearance enhancement was not tested in R3.8–R3.10 because the current model has no aggregation decay term (no $-\text{AGG_CLEARANCE} \times a_i$ component). Future model versions should add explicit clearance kinetics.

2.9 Therapeutic cliff protocol (R3.9)

Nineteen ISR suppression levels from 0% to 99%, with dense sampling near the anticipated threshold. TSSE held constant at 2.0. $N = 50$ seeds $\times$ 500 steps per level = 950 runs. Key derived quantities: (1) cliff detection — $|\Delta \text{genuine_rate}| > 0.20$ or $|\Delta B| > 0.10$ between adjacent levels; (2) ISR_{50} — suppression for 50% reduction in genuine tipping rate, by linear interpolation with bootstrap CI ($N = 1000$); (3) ISR_{90} ; (4) response shape classification.

2.10 Biological mapping protocol (R3.10)

Three interventions tested across 11 strength levels {10%, 20%, ..., 90%, 95%, 99%}:

- **Intervention A — Production suppression:** $\text{ISR} = 2.0 \times (1 - s)$, TSSE fixed at 2.0. Biological analogue: ASO-mediated gene silencing.
- **Intervention B — Spread inhibition:** $\text{TSSE} = 2.0 \times (1 - s)$, ISR fixed at 2.0. Biological analogue: synaptic transmission blockers.

- **Intervention C — Coupled suppression:** both ISR and TSSE = $2.0 \times (1 - s)$ simultaneously. Reproduces v1.0 aggAmp reduction.

$N = 30$ seeds $\times$ 500 steps $\times$ 33 conditions + 30 baseline = 1020 runs. Synergy quantified by **Bliss Independence**: $E_{AB} = A + B - A \times B$, where A, B are fractional benefit scores. Synergy = observed_C - E_{AB} . Bootstrap CI ($N = 1000$ resamples from per-run data; identical seeds for deterministic recovery). Classification: synergistic (> 0.05), antagonistic (< -0.05), additive (± 0.05).

3 Results

3.1 R3.1 — Decoupled aggregation: ISR and TSSE independently load-bearing

Across the 25-cell ISR $\times$ TSSE grid, genuine tipping emerged in 21/25 cells with a mean genuine tipping rate of **0.838**. Tipping is not an artifact of the coupled aggAmp design.

Table 2: Mechanism dominance classification across the 5 $\times$ 5 ISR $\times$ TSSE grid.

Dominant mechanism	Count	Fraction
Seeding (ISR alone sufficient)	10	40%
Spread (TSSE alone sufficient)	4	16%
Both required simultaneously	5	20%
Neither critical (tipping robust to either ablation)	6	24%

Pearson correlation of log(ISR) with genuine tipping rate: $r = 0.586$; for log(TSSE): $r = 0.463$, establishing ISR as the more predictive mechanism. Diagonal cells (ISR = TSSE, comparable to v1.0 aggAmp) reproduce the v1.0 monotonic relationship. The seeding-dominant classification at ISR = TSSE = 2.0 confirms that v1.0 aggAmp dominance reflects **genuine intracellular seeding sensitivity**, not a coupling artifact.

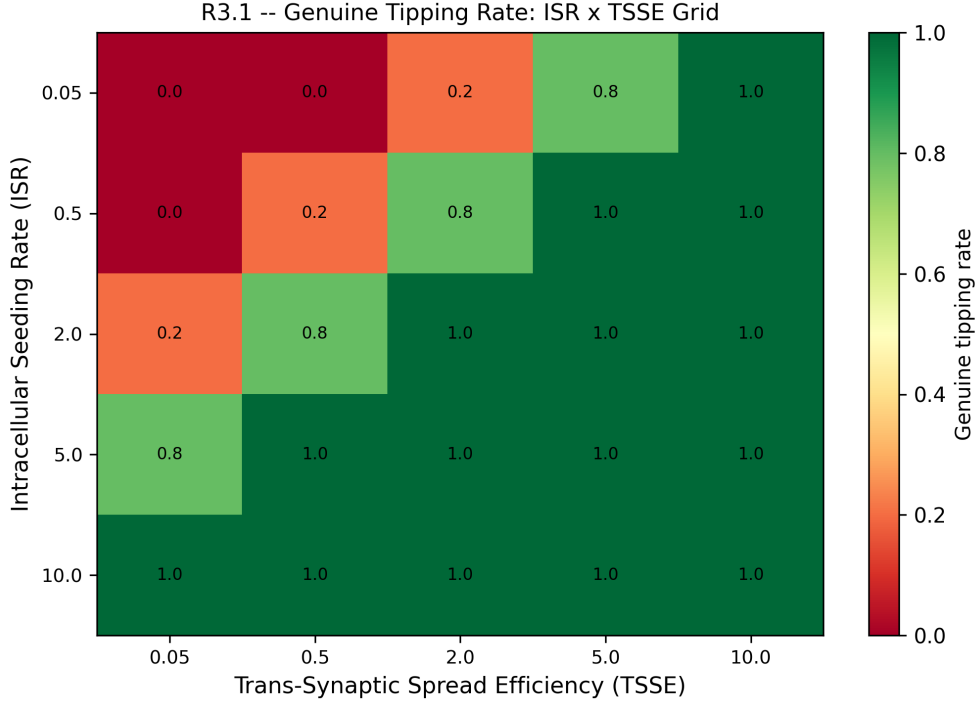


Figure 1: Genuine tipping rate heatmap across the 5×5 ISR $\times$ TSSE parameter grid. Each cell shows the fraction of 20 seeds satisfying the strict Phase 7B tipping criterion. Diagonal cells (ISR = TSSE) correspond directly to v1.0 aggAmp values and reproduce the monotonic relationship. Both row and column gradients confirm independent load-bearing status.

3.2 R3.2 — Downstream causal power: mitochondria conditional, others negligible

Table 3: Mean first-death shift (steps) when each mechanism is ablated, across three dynamical regimes. Effect size in parentheses.

Mechanism	R1: Seeding-dom	R2: Spread-dom	R3: Stressed
ISR (seeding)	+ 179 (large)	+4 (small)	+ 64 (large)
TSSE (spread)	+22 (small)	+ 247 (large)	+33 (medium)
Mitochondria	+3 (small)	+0 (small)	+ 61 (large)
Glutamate	≈ 0 (small)	≈ -2 (small)	+18 (small)
Calcium / ROS	≈ 0 (small)	≈ -2 (small)	+18 (small)
Irreversibility	≈ 0 (small)	≈ -2 (small)	+4 (small)

ISR and TSSE are the sole **universally load-bearing** mechanisms. Mitochondrial damage is **conditionally load-bearing**: large effect only in the downstream-stressed regime (**mitFrag** = 4–8, shift = +61 steps). Glutamate excitotoxicity, calcium/ROS, and irreversibility lock remain negligible across all three regimes.

3.3 R3.3 and R3.4 — Mitochondrial threshold: near-takeover at $\text{mitFrag} = 4.0$

R3.3 ($n = 15$ seeds) initially detected a TAKEOVER at $\text{mitFrag} = 0.3$ in the low-aggregation context via a single criterion hit. R3.4 validated with $n = 50$ seeds and bootstrapped CIs:

Table 4: R3.4 bootstrap-validated mitochondrial threshold analysis (low-aggregation context, $n = 50$ seeds). Bold row = genuine near-takeover onset.

mitFrag	Shift [95% CI]	Gain [95% CI]	rdrop [95% CI]	Criteria
0.3	+1 [−7, +9]	+0.9 [−0.5, +2.4]	−0.060 [−0.24, +0.14]	0/3
3.0	+44 [+36, +51]	+4.9 [+3.5, +6.3]	−0.020 [−0.18, +0.14]	0/3
4.0	+66 [+60, +72]	+7.9 [+6.7, +9.1]	−0.040 [−0.20, +0.12]	2/3
8.0	+106 [+98, +113]	+10.8 [+9.7, +11.9]	+0.060 [−0.06, +0.18]	2/3

The $\text{mitFrag} = 0.3$ signal from R3.3 is a **confirmed artifact** (all three criteria fail at $n = 50$; CIs span zero). The genuine signal begins at $\text{mitFrag} = 4.0$ (shift +66 steps, CI [+60, +72] entirely above the 50-step threshold). The genuine-rate drop criterion is not satisfied at any level: mitochondria function as a *death accelerator* (earlier onset, fewer plateau survivors) but not a *cascade gatekeeper*.

3.4 R3.5 — Topological necessity: ISR topology-invariant, TSSE topology-sensitive

Table 5: Topology comparison: genuine tipping rate and spatial coherence under decoupled model (v2.0) vs. v1.0, at medium context.

Topology	Genuine rate (v2.0)	Genuine rate (v1.0)	Coherence r
<i>C. elegans</i> (biological)	1.000	1.000	0.668
Degree-preserved shuffle	0.895	0.730	0.466
Watts–Strogatz (small-world)	0.870	0.780	0.520
Erdős–Rényi (random)	0.700	0.480	0.396
Barabási–Albert (scale-free)	0.105	0.000	0.073

The decisive finding is the mechanism-isolation split:

Table 6: Genuine tipping rate under ISR-dominant vs. TSSE-dominant conditions across all five topologies.

Topology	ISR-dom (ISR=5, TSSE=0.5)	TSSE-dom (ISR=0.5, TSSE=5)
<i>C. elegans</i>	1.000	1.000
Erdős–Rényi	1.000	0.000
Barabási–Albert	1.000	0.000
Watts–Strogatz	1.000	0.050
Degree-preserved shuffle	1.000	0.000

Under ISR-dominant conditions, **all five topologies achieve 100% genuine tipping**. Under TSSE-dominant conditions, **only the biological *C. elegans* connectome tips reliably**; all four synthetic alternatives fail (0–5%). The entire topology sensitivity of the cascade is carried by the trans-synaptic spread mechanism.

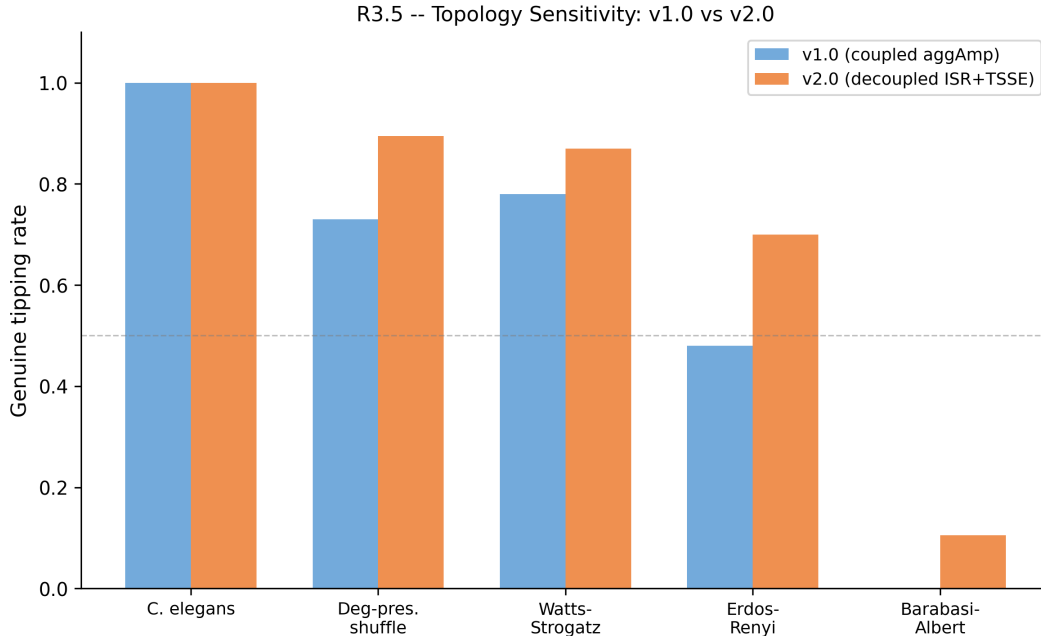


Figure 2: Genuine tipping rate across five topology types under (left) ISR-dominant and (right) TSSE-dominant conditions. ISR-dominant achieves 100% on all topologies; TSSE-dominant achieves 100% only on the biological connectome.

3.5 R3.6 — Seed location: timing modulator, not tipping determinant

At medium context ($\text{ISR} = \text{TSSE} = 2.0$), genuine tipping rate = 1.000 for 59/61 neurons (exceptions: PLML and PLMR, peripheral sensory neurons with vulnerability = 0.05). The parameter regime, not seed location, determines whether tipping occurs.

Table 7: Named-condition seed location comparison at medium context.

Condition	Genuine rate	First death step	Plateau survivors
AVAL boost (command hub, rank 9/61)	1.000	77	9.6
DA6 boost (protective node, rank 19)	1.000	80	9.2
Uniform baseline (no focal seed)	1.000	165	9.2

The onset timing range is **88 steps** (77–165), representing 17.6% of the 500-step window. Spearman $r(\text{vulnerability}, \text{onset speed}) = 0.612$, confirming vulnerability as a moderate predictor while establishing network position as an independent predictor.

3.6 R3.7 — Degree-correlated vulnerability: BA failure reflects misalignment

Table 8: Effect of vulnerability assignment on genuine tipping rate and spatial coherence under TSSE-dominant conditions. Deg-death corr = Spearman r between node degree and death step.

Topology	Vulnerability	Genuine rate	Coherence r	Deg-death corr
BA	A: Original	0.000	−0.265	0.411
BA	B: Degree-correlated	0.800	+0.371	0.401
BA	C: Inverse-degree	0.000	−0.326	0.419
<i>C. elegans</i>	A: Original	1.000	+0.453	0.289
<i>C. elegans</i>	B: Degree-correlated	0.000	+0.278	0.408
<i>C. elegans</i>	C: Inverse-degree	0.000	−0.291	0.482

The degree-death correlation (≈ 0.41) is stable across all BA conditions: TSSE always routes spread through hubs first. Under degree-correlated vulnerability (B), BA genuine tipping rises from 0.000 to **0.800** and coherence flips from -0.265 to $+0.371$. Conversely, reassigning *C. elegans* to degree-correlated vulnerability destroys tipping entirely ($1.000 \rightarrow 0.000$). **The R3.5 BA failure is attributable to hub-vulnerability misalignment, not hub architecture per se.**

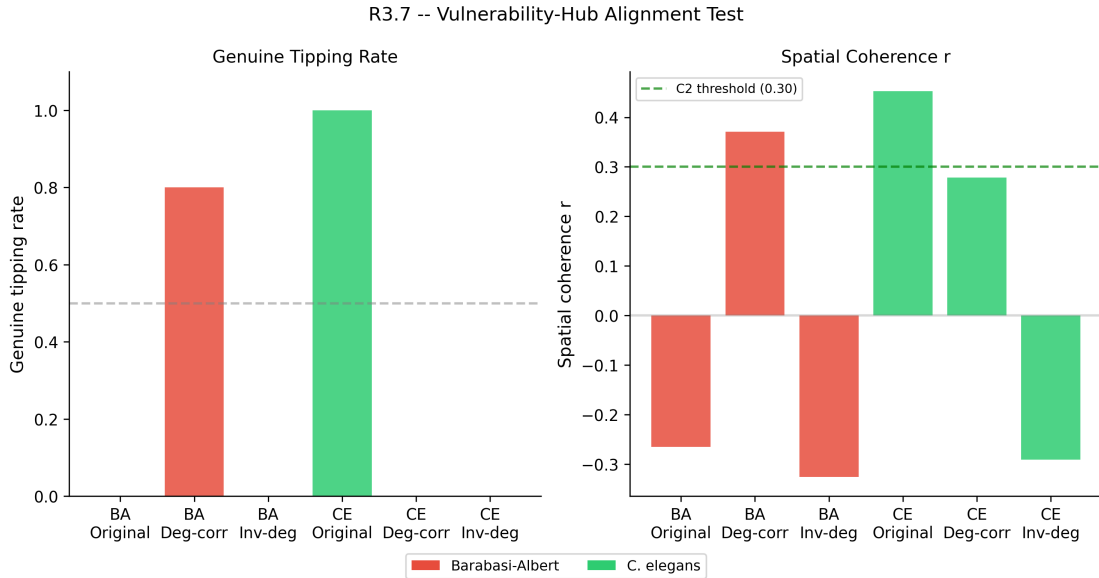


Figure 3: Effect of vulnerability assignment on TSSE-driven cascade under BA and *C. elegans* topologies. Degree-correlated assignment (B) rescues BA from 0% to 80%; the same assignment destroys *C. elegans* ($100\% \rightarrow 0\%$). The symmetric result confirms that the *C. elegans* motor circuit’s success depends on hub-vulnerability alignment, not topology statistics.

3.7 R3.8 — Combination therapy: ISR is cascade gatekeeper, glutamate negligible

Under medium context, all 22 tested conditions produced genuine tipping at 100% when ISR suppression $\leq 60\%$. Glutamate suppression at any dose (50%, 70%, 90%, 99%) left genuine tipping rate unchanged at 1.000.

Table 9: Key quantitative comparison at matched dose levels (R3.8).

Intervention	Dose	Genuine tipping	Benefit score
ISR-only	50%	1.000	0.057
Glutamate-only	90%	1.000	0.001
Ratio (ISR / glut at comparable doses)	—	—	63×

Glutamate suppression is 63-fold less effective than ISR suppression. Even at 99% glutamate suppression, genuine tipping rate remains 1.000 and benefit score is 0.0003. All 12 combination pairs were classified as **additive** (all bootstrap CIs overlapped zero). No combination provided benefit beyond the ISR component alone.

Verdict: GATEKEEPER-DOMINANT. ISR controls cascade entry; glutamate acts downstream of the bottleneck and cannot prevent tipping regardless of suppression depth.

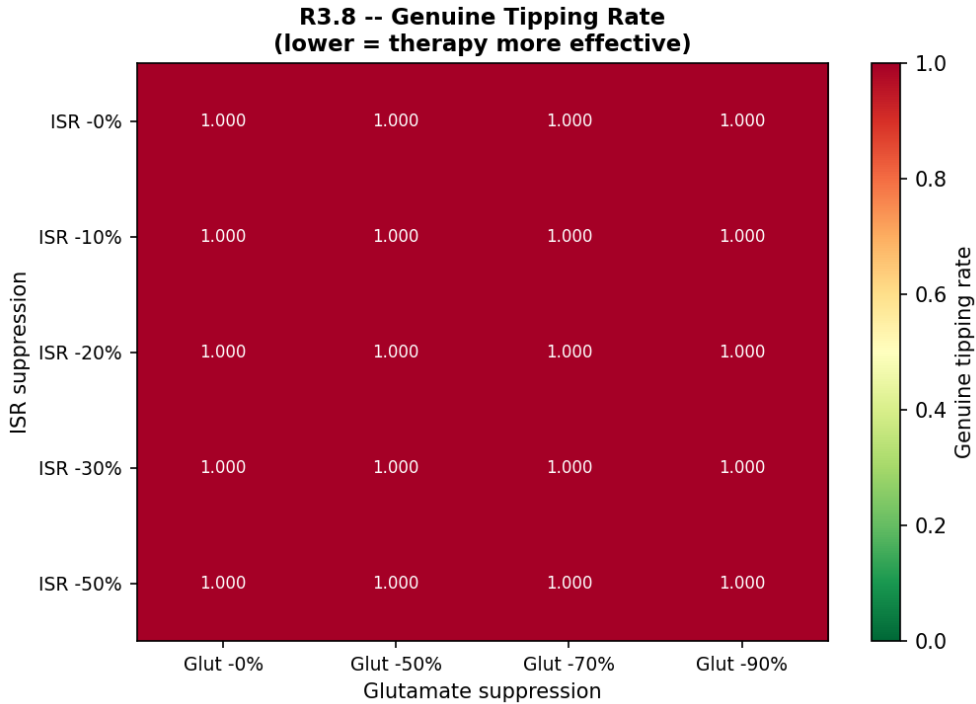


Figure 4: R3.8 — Genuine tipping rate across the 5×4 ISR $\times$ glutamate suppression grid. Each cell shows the fraction of 20 seeds satisfying the Phase 7B criterion. Tipping remains at 100% for all cells with ISR suppression $\leq 60\%$, regardless of glutamate dose. Genuine tipping first falls below 100% only in the top rows (ISR $\geq 70\%$).

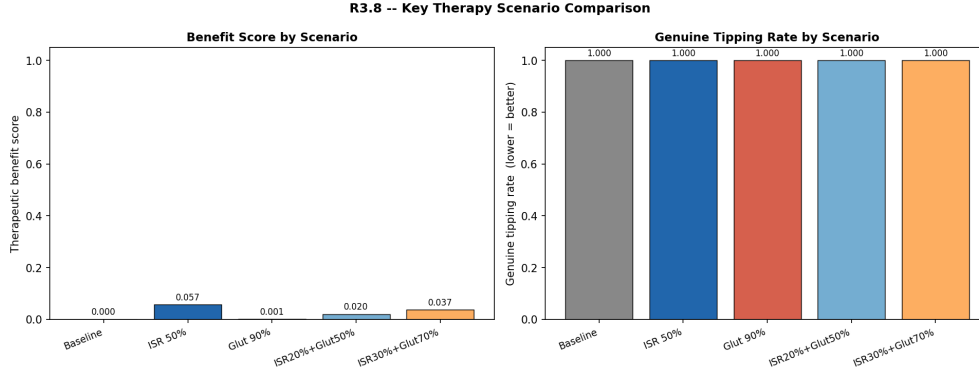


Figure 5: R3.8 — Benefit score for key therapy scenarios. ISR suppression (blue) dominates all glutamate-only (orange) and combination conditions. The $63\times$ benefit ratio between ISR 50% and glutamate 90% is visible directly.

3.8 R3.9 — Therapeutic cliff: dose-response is threshold-gated

The 19-level ISR suppression sweep (0%–99%, $N = 950$ runs) reveals a cliff-like dose-response. Between 85% and 90% suppression:

Table 10: Cliff detection at 85%–90% ISR suppression (R3.9). Values in brackets are 95% bootstrap CIs.

Metric	85% suppression	90% suppression	Step change
Genuine tipping rate	0.860 [0.76, 0.94]	0.580 [0.44, 0.70]	− 0.280
Benefit score	0.163	0.293	+ 0.131

This 28-percentage-point single-step drop exceeds the cliff threshold ($|\Delta| > 0.20$). $\text{ISR}_{50} = 96.5\%$ [95% CI: 89.3%, 97.3%]. ISR_{90} was *not reached* at 99% suppression (genuine tipping at 99% = 0.52 [0.38, 0.66]). Residual tipping is sustained by TSSE = 2.0, held constant throughout R3.9. **Response shape classification: CLIFF-LIKE.**

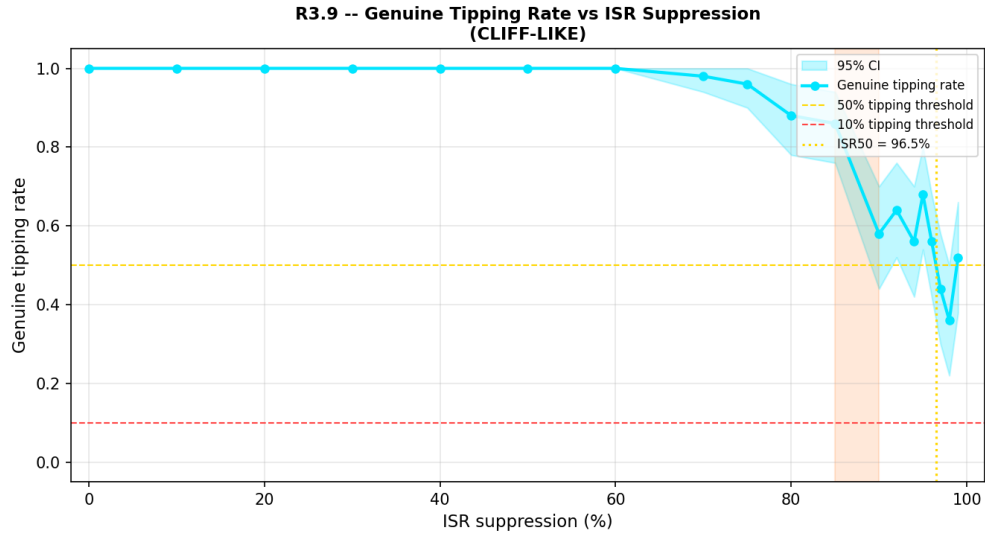


Figure 6: R3.9 — Genuine tipping rate vs. ISR suppression level (19 levels, $N = 50$ seeds each). The steep drop between 85% and 90% is the sole cliff across the full dose-response curve. Shaded band: 95% bootstrap CI. Dashed line indicates $\text{ISR}_{50} = 96.5\%$.

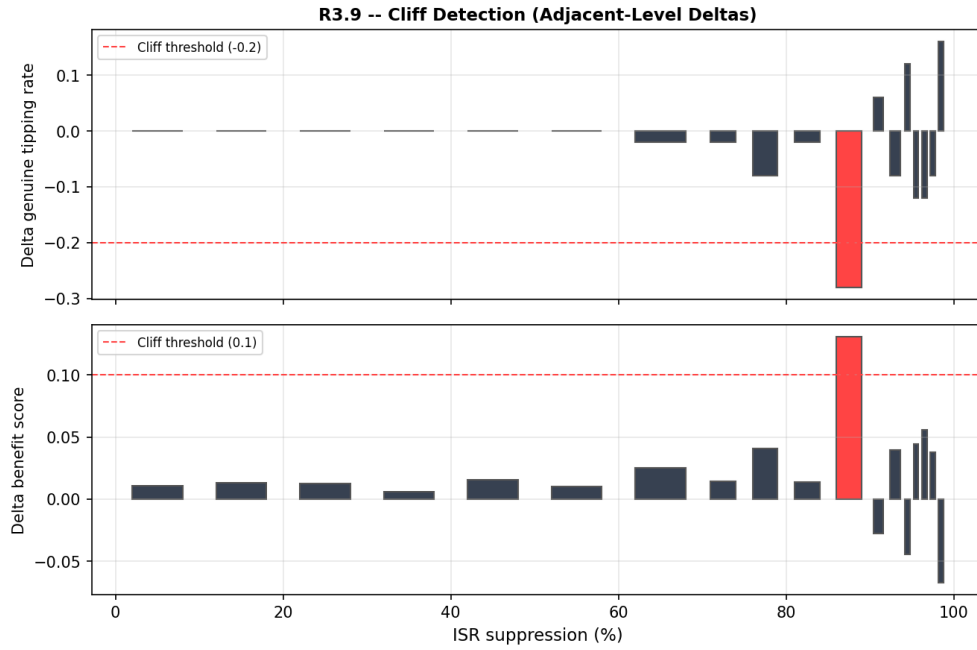


Figure 7: R3.9 — Step-wise change in genuine tipping rate between adjacent suppression levels. The single exceedance of the cliff threshold ($|\Delta| > 0.20$) at the 85%→90% transition confirms the CLIFF-LIKE classification.

3.9 R3.10 — Biological mapping: production suppression dominates; synergy above 60%

Table 11: Three-intervention summary at 30 seeds per level across 11 strength levels. ISR_{50} = strength at which genuine tipping rate falls to 50%.

Intervention	Genuine tip @ 99%	Best benefit	ISR_{50} strength	Cliff
A: Production only (ASO analogue)	0.500 [0.33, 0.67]	0.342	95%	90%→95%
B: Spread only (synaptic blocker)	1.000	0.137	Not reached	None
C: Coupled (v1.0-style)	0.000	0.979	72%	60%→70%

Intervention B cannot prevent genuine tipping at any tested strength. Genuine tipping rate = 1.000 across all 11 conditions from 10% to 99% TSSE suppression. Coupled suppression (C) achieves near-complete prevention above 90% (genuine tipping = 0.000 [0.00, 0.00]).

Table 12: Bliss Independence synergy analysis at key strength levels (R3.10). $E_{AB} = A + B - A \times B$; Synergy = $\text{Observed}_C - E_{AB}$.

Strength	Ben_A	Ben_B	Bliss E_{AB}	Ben_C	Synergy	95% CI
10–50%	(incremental, CIs all overlap zero)				−0.004 to +0.030	<i>Additive</i>
60%	0.072	0.061	0.129	0.183	+0.055	[+0.044, +0.067]
70%	0.104	0.072	0.168	0.416	+0.247	[+0.166, +0.322]
80%	0.186	0.083	0.253	0.719	+0.466	[+0.381, +0.535]
90%	0.191	0.102	0.273	0.904	+0.631	[+0.584, +0.672]

Mean Bliss synergy across all 11 strength levels: benefit = +0.229, tipping protection = +0.258. Six of 11 strength levels show statistically significant synergy ($\text{CI}_{\text{lo}} > 0$). **Synergy onset coincides exactly with the R3.9 therapeutic cliff** (60–70% strength). The synergy is mechanistic cooperativity: ISR suppression creates the context required for TSSE co-suppression to be effective, not pharmacological synergy between two independently active agents.

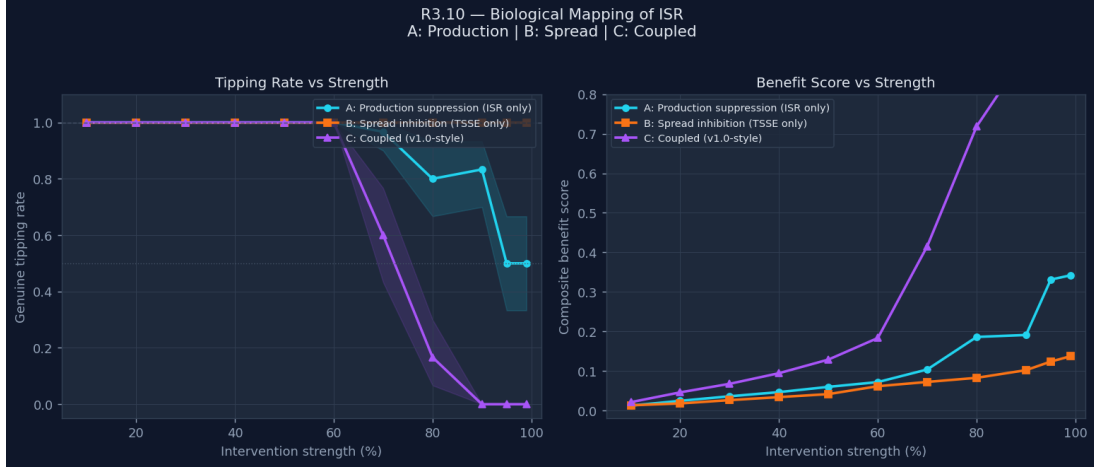


Figure 8: (Left) Genuine tipping rate vs. intervention strength for production suppression (A, cyan), spread inhibition (B, orange), and coupled co-suppression (C, purple). Spread inhibition alone cannot reduce tipping below 100% at any tested dose. (Right) Bliss synergy score per strength level; green bars indicate statistically significant synergy ($CI_{lo} > 0$); onset aligns with the R3.9 therapeutic cliff at 60–70%.

4 Discussion

4.1 ISR as cell-autonomous disease mechanism

The topology-invariance of ISR-dominant cascades (100% genuine tipping across all five graph types) has direct mechanistic implications. Intracellular seeding does not require any specific circuit architecture to drive tipping, consistent with the cell-autonomous biology of TDP-43 aggregation: protein misfolding is initiated within individual neurons by local factors (RNA-binding protein dysfunction, nuclear-cytoplasmic transport failure) independent of synaptic connectivity. The comparative predictiveness of ISR over TSSE ($r = 0.586$ vs. 0.463) is consistent with this view.

4.2 TSSE as circuit-dependent disease mechanism

The catastrophic topology-sensitivity of TSSE-dominant cascades (100% on *C. elegans*, 0–5% on four synthetic alternatives) establishes trans-synaptic spread as a mechanism requiring vulnerability-ordered circuit architecture. R3.7 identifies the precise requirement: TSSE-driven spread always routes through high-degree hub nodes first (degree-death correlation ≈ 0.41 in BA, stable across all vulnerability assignments), but spatial coherence (C2 criterion) emerges only when this hub-first death order aligns with the vulnerability gradient.

The *C. elegans* motor circuit achieves this alignment biologically: motor neurons (DA, DB, DD, VA, VD) are both the most synaptically connected nodes and the most vulnerable to aggregation stress. This refines the R3.5 claim from “*TSSE requires specific circuit architecture*” to “*TSSE requires that the circuit’s hub structure aligns with its vulnerability gradient*” — a testable hypothesis using single-cell transcriptomics combined with connectome reconstruction.

4.3 Mitochondria as death accelerator, not cascade gatekeeper

The validated near-takeover threshold at $\text{mitFrag} \geq 4.0$ reveals that mitochondrial damage accelerates death onset (+66 steps, CI [+60, +72]) and suppresses plateau survival (+7.9 neurons) but does not disrupt tipping structure. This positions mitochondrial dysfunction as a **disease modifier** rather than disease initiator in this model — consistent with the modest efficacy observed for mitochondrial-targeted approaches in clinical ALS trials.

4.4 Implications for v1.0 claims

Round 3 strengthens several v1.0 claims while qualifying others:

1. **aggAmp dominance is genuine** (R3.1): The v1.0 finding is not a coupling artifact.
2. **Topology dependence is TSSE-specific** (R3.5): Resolves Phase 7C ambiguity.
3. **Single-factor claim qualified** (R3.2/R3.4): Mitochondria acquire near-takeover character at extreme fragility ($\geq 4\times$ baseline) in low-aggregation contexts.
4. **Seed location not primary** (R3.6): Validates Phase 1A directionally; the broader circuit is broadly susceptible (59/61 neurons fragile) at medium intensity.
5. **Topology sensitivity is alignment-dependent** (R3.7): BA hub structure always drives hub-first spread; resistance arises from misalignment of hub status and vulnerability.

4.5 Therapeutic implications of R3.8–R3.10

Glutamate excitotoxicity is a poor primary target. R3.8 demonstrates that glutamate suppression at any dose does not reduce genuine tipping. The glutamate pathway is downstream of ATP depletion, which itself is downstream of aggregation. This is consistent with the observation that riluzole (which reduces glutamate release) provides only modest benefit in ALS [Benatar et al., 2023], though the model-to-biology mapping is approximate.

Near-complete ISR engagement is required. The cliff-like dose-response ($\text{ISR}_{50} = 96.5\%$) implies that partial ISR suppression produces only incremental benefit. If the ISR analogy to TDP-43 expression holds, ASO studies should test whether transcript reduction and aggregation prevention are similarly non-linear.

The synergy is mechanistic cooperativity, not additive independence. Intervention B alone provides zero tipping protection, so the Bliss expected combined = prot_A at all doses. Any coupling gain registers as synergy, reflecting genuine cooperative cascade dynamics: ISR suppression creates the conditions under which TSSE co-suppression can function. This is analogous to a sensitiser + effector combination rather than two independently active drugs.

Model limitation. The current model lacks an aggregation decay term; clearance-enhancing interventions (autophagy inducers, proteasome activators) cannot be evaluated. Adding a $-\text{AGG_CLEARANCE} \times \text{atp} \times a_i$ term would enable a third intervention class and allow comparison of production suppression, spread inhibition, and clearance enhancement as distinct biological strategies.

4.6 Limitations of the mechanistic interpretation

Topology-invariance of ISR and topology-sensitivity of TSSE are model properties that may not map directly onto biological mechanisms. Vulnerability scores, synapse weights, and neurotransmitter assignments are approximate. The discrete 5×5 parameter sweep cannot rule out non-monotonic behaviour between grid points. The R3.7 BA contingency is resolved but whether hub-vulnerability alignment generalises beyond *C. elegans* remains untested. R3.6 tested seed sensitivity at medium aggregation only.

5 Conclusion

Decoupling aggregation into intracellular seeding (ISR) and trans-synaptic spread (TSSE) resolves a key mechanism identifiability problem in the v1.0 model. Together with three therapeutic-mapping phases, these yield eight principal findings:

1. **aggAmp dominance is genuine:** ISR and TSSE are independently load-bearing, with ISR more predictive ($r = 0.586$ vs. 0.463). Not a coupling artifact.
2. **ISR is topology-invariant:** Under ISR-dominant conditions, all five tested topologies achieve 100% genuine tipping. Intracellular seeding is cell-autonomous.
3. **TSSE is topology-sensitive via vulnerability-hub alignment:** Only the biological *C. elegans* connectome produces genuine TSSE-dominant tipping (100%); all four synthetic alternatives fail (0–5%). Degree-correlated vulnerability rescues BA to 80%; the same assignment collapses *C. elegans* from 100% to 0%.
4. **Mitochondria: death accelerator, not gatekeeper:** At $\text{mitFrag} \geq 4.0$, mitochondrial damage accelerates onset (+66 steps) and suppresses survival (+7.9 neurons) but does not gate tipping. Glutamate, calcium/ROS, and irreversibility remain negligible across all regimes.
5. **Seed location is a timing modulator:** 59/61 neurons produce genuine tipping regardless of boost location. The seed location affects *when* (88-step range) but not *whether* tipping occurs.
6. **Glutamate suppression 63-fold less effective than ISR:** Even at 99% glutamate reduction, genuine tipping rate = 1.000. All ISR $\times$ glutamate combination synergies are additive (R3.8).
7. **ISR dose-response is cliff-like:** The 85–90% ISR suppression interval produces a 28-pp single-step drop — the only cliff across the full 0–99% sweep. $\text{ISR}_{50} = 96.5\%$ [89.3%, 97.3%]; ISR_{90} not reached at 99% (R3.9).
8. **Production suppression $\gg$ spread inhibition; coupled suppression synergistic above 60%:** Spread inhibition alone cannot prevent tipping at any dose. Production suppression alone achieves 50% tipping reduction at 95% strength. Coupled co-suppression achieves 100% prevention above 90% strength, with Bliss synergy +0.47 at 80% (CI [+0.38, +0.53]), onset coinciding exactly with the R3.9 cliff (R3.10).

6 Limitations

1. **Organism gap:** The *C. elegans* motor connectome (61 neurons, 127 synapses) is orders of magnitude simpler than the human corticospinal-motor neuron system. Findings are specific to this substrate.
2. **Parameter ranges:** ISR and TSSE are dimensionless scaling factors, not calibrated to measured biological quantities.
3. **Single ablation:** The R3.2 protocol disables one mechanism at a time. Simultaneous multi-pathway ablations may reveal non-additive dependencies.
4. **Discrete topology types:** The five graph types are mathematical idealisations. Whether hub-vulnerability alignment generalises beyond *C. elegans* is untested.
5. **Seed location at medium aggregation only:** At lower aggregation intensity, seed location may be a stronger determinant of tipping outcome.
6. **No aggregation clearance term:** Clearance-enhancing interventions cannot be modelled. Adding explicit clearance kinetics would enable a third intervention class.
7. **All results hypothesis-generating:** Not a clinical predictor of ALS progression, treatment response, or patient prognosis.

Data Availability

The complete simulation engine, phase-by-phase results (JSON), and this manuscript source are available at <https://github.com/MaryamRezaeeN1/als-connectome>.

Zenodo archive v1.0 (DOI): <https://doi.org/10.5281/zenodo.20528826>

Zenodo archive v2.0 (DOI): <https://doi.org/10.5281/zenodo.20592681>

An interactive web-based visualisation is available at <https://als-connectome-dashboard.vercel.app>.

References

- Michael Benatar et al. Randomized, double-blind, placebo-controlled trial of tofersen in presymptomatic SOD1 variant carriers (ATLAS). *Annals of Neurology*, 93(5):974–988, 2023. doi: 10.1002/ana.26649.
- Steven J. Cook, Travis A. Jarrell, Christopher A. Bhatt, Maryam Schmidt, Abhishek Bhatt, et al. Whole-animal connectomes of both *Caenorhabditis elegans* sexes. *Nature*, 571:63–71, 2019. doi: 10.1038/s41586-019-1352-7.
- Manuela Neumann, Deepak M. Sampathu, Linda K. Kwong, et al. Ubiquitinated TDP-43 in frontotemporal lobar degeneration and amyotrophic lateral sclerosis. *Science*, 314(5796):130–133, 2006. doi: 10.1126/science.1134108.
- Maryam Rezaee. Cascade criticality, therapeutic window emergence, and pre-symptomatic early-warning signals in a connectome-based neurodegeneration framework: an als-motivated study on the *C. elegans* motor circuit, 2026. URL <https://doi.org/10.5281/zenodo.20528826>. v1.0.

John G. White, Eileen Southgate, J. Nichol Thomson, and Sydney Brenner. The structure of the nervous system of the nematode *Caenorhabditis elegans*. *Philosophical Transactions of the Royal Society B*, 314(1165):1–340, 1986. doi: 10.1098/rstb.1986.0056.